A Dual-Stage Residual Diffusion Model With Perceptual Decoding for Remote Sensing Image Dehazing

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***Abstract*—Atmospheric pollutants, such as haze, severely affect the quality of remote sensing images, leading to blurred details and impairing their effectiveness in applications like environmental monitoring and agricultural resource management. In recent years, diffusion models have attracted widespread attention due to their powerful generative capabilities. However, striking a balance between their expensive training costs and actual recovery effectiveness has become a major challenge. To address this key challenge, we propose a dual-stage residual diffusion model with perceptual decoding (DS-RDMPD) for remote sensing image dehazing. The core innovation of our work lies in a DS coarse-to-fine architecture that integrates the traditional U-Net with a diffusion model, enabling efficient adaptation of diffusion-based restoration to the dehazing task. This design not only achieves strong performance but also demonstrates remarkable generalization across various image restoration scenarios. In the first stage, we use the multichannel efficient selective synthesis U-Net (MCESS-UNet) to preprocess the remote sensing haze images. This architecture performs initial dehazing and feature extraction through a multiscale channel attention (MC) block and then performs enhanced spatial feature aggregation through an efficient selective synthesis (ESS) block. The preprocessed image is then used as the conditional input of the RDMPD, where the perceptual decoder improves the generation quality by further decoupling the condition to refine the residual estimate. Extensive experiments on multiple datasets show that DS-RDMPD can achieve satisfactory results with only 300 000 iterations and about five sampling steps. It has achieved satisfactory results in both qualitative and quantitative experiments, and also exhibits strong performance in rain removal and deblurring tasks, demonstrating the excellent generalization ability of the model. The code is available at https://github.com/Aaronwangz/DS-RDMPD**

***Index Terms*—Computer vision, diffusion model, multiscale channel attention (CA), remote sensing image dehazing.**

Received 28 April 2025; revised 10 July 2025; accepted 11 August 2025. Date of publication 19 August 2025; date of current version 9 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62401012, in part by the Collaborative Innovation Project of Anhui Universities under Grant GXXT-2023-021, and in part by the Key Project of Natural Science Foundation of Anhui Provincial Department of Education under Grant 2022AH050319. *(Corresponding author: Tao Tao.)*

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Digital Object Identifier 10.1109/TGRS.2025.3600540

I. Introduction

REMOTE sensing image dehazing is a critical task in applications such as environmental monitoring and disaster assessment. Atmospheric particulate matter (e.g., haze, clouds, and rain) can significantly degrade image quality and negatively affect high-level visual tasks such as object detection and scene classification. Traditional dehazing methods mainly rely on physical models [1] or empirical priors [2], but they often fail to perform well under complex and nonuniform haze conditions and struggle to handle diverse real-world scenarios. With the rise of deep learning, convolutional neural networks (CNNs) [3], [4], [5] have advanced remote sensing image dehazing by effectively extracting deep features. However, CNNs have difficulty in capturing long-range dependencies. To address this problem, Transformer models [6], [7], [8] have been introduced, as they are more effective at modeling global context. However, their high computational cost limits their practical applications. Unlike CNN and Transformer methods that rely on feature extraction, generative models (such as GANs and diffusion models [9]) directly generate dehazed images by learning the latent distribution of the input. In contrast to traditional methods that focus on feature restoration, generative models are capable of reconstructing images either from degraded inputs or even using textual prompts for guidance. Owing to their more stable training process compared to GANs, diffusion models have attracted increasing attention in recent years. They have demonstrated outstanding performance in image restoration tasks and have gradually become a major focus of research in the field of image dehazing.

Recently, RSHazeDiff [10] has emerged as one of the most advanced remote sensing dehazing techniques based on diffusion models. It leverages the Fourier transform to capture the global dependencies of the input image and incorporates a coarse-to-fine training strategy to regulate the diffusion model's training process. Despite its effectiveness, RSHazeDiff, like many other image restoration methods based on diffusion models, suffers from the need for millions of iterations [11], [12], [13] and excessively long sampling steps. To address this limitation, the residual denoising diffusion model (RDDM) [14] was introduced. RDDM innovatively adopts a double diffusion process that decomposes the denoising task into residual diffusion and noise diffusion components.

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Fig. 1. (a) Haze image. Result of (b) RDDM and (c) our model. (d) Ground truth, demonstrating the impact of the model under extreme haze conditions.

This approach improves determinism, prioritizing the residual component for deterministic behavior, while the noise component emphasizes diversity. RDDM effectively reduces the required training resources and sampling steps by optimizing this balance. However, as demonstrated in Fig. 1, the dehazing performance of RDDM [14] remains limited when applied to severely degraded images.

We believe that variations in haze density make recovery difficult, a challenge that is particularly pronounced in remote sensing images. Due to varying atmospheric thickness, remote sensing images often exhibit large-scale spatial heterogeneity and uneven haze distributions, making restoration particularly challenging. These challenges limit the effectiveness of end-to-end networks, which often struggle to recover global semantic structures and fine spatial details simultaneously. With this in mind, we propose a dual-stage residual diffusion model with perceptual decoding (DS-RDMPD). Our approach adopts a DS architecture: the first stage performs coarse dehazing to restore the overall scene structure, while the second stage refines residual errors under perceptual guidance. This strategy effectively enhances dehazing performance while reducing computational overhead.

Specifically, in the first stage, two blocks are implemented to handle feature extraction and fusion selection. The multiscale channel attention (MC) block applies convolutional layers at different scales to capture diverse features, aiding in the identification of haze patterns and enhancing the model's generalization. Furthermore, we propose the efficient selective synthesis (ESS) block, which replaces traditional skip connections and selectively fuses the extracted features with weighted emphasis. Leveraging these two blocks, we develop the multichannel ESS U-Net (MCESS-UNet) model as part of the coarse dehazing process to generate the preprocessed image.

In the second stage, the preprocessed image is input into the diffusion model for further optimization. Unlike traditional RDDMs, the proposed DS-RDMPD framework incorporates a conditional feature fusion (CFF) module. This module enables the encoder to dynamically perceive and integrate conditional information, facilitating a more effective decoupling of such information. This approach addresses the limitations of conventional diffusion models, which typically rely on simple upsampling operations that overemphasize the encoder's output features and consequently lead to the loss of important details. The inclusion of the CFF module improves the guidance of image generation, offering valuable insights for future research. Overall, compared to conventional image dehazing diffusion models, DS-RDMPD significantly reduces both training time and sampling steps. Our DS-RDMPD model achieves high-quality results with approximately 300 000 training steps from scratch and only five sampling steps, effectively balancing dehazing performance and computational efficiency. The main contributions of this article are as follows.

1) We propose a novel DS remote sensing image dehazing network that combines the advantages of U-Net and diffusion model, achieving satisfactory results without requiring millions of training iterations or excessive generation time.
2) In the first stage, we design MCESS-UNet, a U-Net variant tailored to remote sensing images, featuring a MC block and an ESS block. These modules enhance multiscale feature extraction and adaptive fusion, improving the quality of initial estimates for downstream refinement.
3) In the second stage, we design a new perceptual decoding method that effectively integrates and decouples preprocessed images from degraded images, further enhancing the recovery capability of the RDMPD and bridging the gap between haze images and clear images.
4) The experimental results demonstrate that DS-RDMPD outperforms most existing methods in remote sensing image dehazing. Additionally, it achieves promising results across multiple image restoration tasks, including rain removal and deblurring, highlighting its strong generalization capabilities. This provides significant performance improvements and effective solutions for practical applications.

II. Related Work

A. Remote Sensing Dehazing

In recent years, some remote sensing dehazing methods have been based on the combination of physical models and neural networks. Among them, DehazeNet [3] was the first method to combine the two and achieved the best results at that time. In addition, AOD-Net [4] redefined the physical model and combined it with a neural network. However, the model built based on the physical scattering model equation is a highly ill-conditioned problem due to the unknown transmission map and global atmospheric light, and it is difficult to cope with various practical scenarios.

In contrast, variant structures of specialized U-Net architectures constructed by CNN or Transformer [15], [16], [17], [18], [19], [20], [21], [22], [23], [24] have been widely used. Among them, Trinity-Net [7] was a remote sensing image dehazing model based on the Swin Transformer, which aims to restore the real surface information by integrating prior knowledge and deep learning strategies. Wen et al. [5] proposed EMPF-Net, which combines physical priors with a CNN to achieve high-quality results with fewer parameters. DehazeFormer [6] proposed a new method with an improved normalization layer and a spatial aggregation scheme. PCSformer [8] was a novel Transformer method for remote sensing image dehazing. By introducing sliding cross-stripe Transformer blocks and local proxy-based global Transformer blocks, it could efficiently

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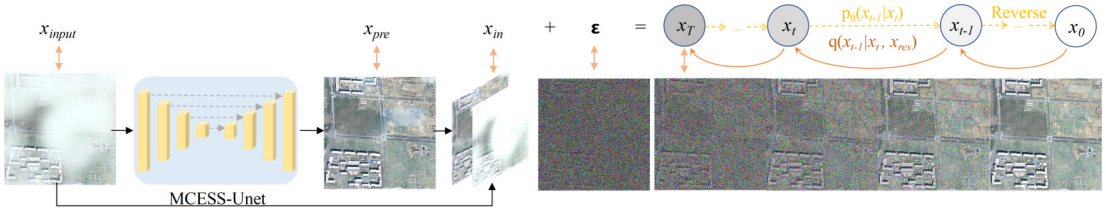


Fig. 2. Schematic of the developed DS-RDMPD. First, the hazy image (x_{input}) is processed by the MCESS-UNet to generate a preprocessed image (x_{pre}). Next, x_{pre} is concatenated with the hazy image to form the input (x_{in}), which serves as a condition to predict the residual. Subsequently, by generating a degraded version x_T of x_{pre}, the RDMPD is utilized to efficiently restore a clear image. The entire framework first uses the feature extraction capability of MCESS-UNet to perform coarse dehazing and then uses RDMPD to restore fine details.

model long-range dependencies and restore details of dense haze scenes. This method addressed the limitations of existing methods while maintaining low computational complexity and reducing artifacts through a shallow-depth model.

Li et al. [25] proposed a lightweight remote sensing image dehazing method, the progressive self-optimizing network (PSR Net). This method used a pre-dehazing module and a progressive self-optimizing module, combined with a restoration attention module (RAB) to iteratively optimize the dehazing effect, and designed a spatial adaptive feature extraction module to dynamically adjust the receptive field. To address the problems of irregular haze shapes and uneven concentrations, KFA-Net [26] introduced asymmetric scale feature cascade (ASFC), K-means pixel attention (KPA), and FFT channel attention (FCA) mechanisms to optimize feature extraction, haze area focusing, and frequency domain feature acquisition. Liang et al. [27] proposed a remote sensing image dehazing method based on heterogeneous priors, extracting superpixels through the DSLIC algorithm, using the ALFE model to estimate atmospheric light, combining HP-HCM to derive global atmospheric light values, and introducing the DDIP model to refine the transmission map. Although this article shows that this method is computationally more efficient than other methods, it still uses a large number of parameters and expert modules, which may pose certain challenges in resource-constrained environments. Sun et al. [28] proposed a partial twin multiscale dual codec dehazing network (PSMB-Net), which constructed dehazing prior information through a partial twin framework (PSF) to guide the network to optimize the dehazing effect, and used a multiscale dual codec information fusion (MBIF) module to improve color and texture reconstruction capabilities. Combining the advantages of physical models and neural networks, Du et al. [29] proposed AU-Net, which used an atmospheric scattering model to generate a rough dehazed image by jointly optimizing the atmospheric light (A) and the transmission map (T) and improving the image quality through a refinement module. The network combines a deep self-attention (SAD) module and a channel attention (CA) module to enhance the fusion of deep semantics and channel information, respectively. However, our reproduction results of the model show that the generalization ability of the model is limited.

B. Diffusion Model

In recent years, diffusion models [30], [31] have shown great potential in image dehazing, but there have still been few studies on remote sensing image dehazing. DehazeDDPM [12] combined an atmospheric scattering model with conditional diffusion to improve performance, but its long sampling steps required significant time. WeatherDiffusion [11] was a groundbreaking diffusion-based method for image restoration. It could handle images of various sizes, but the sampling step was not optimized, and the inference time was long due to the large number of image patches. In addition, WaveDM [13] built upon WeatherDiffusion [11] by integrating the wavelet transform and introducing new optimizations for the sampling process, which sped up image generation. Recently, RSHazeDiff [10] used the Fourier transform to capture frequency-domain information and preserve more detailed textures. However, the above diffusion models require at least two million training iterations to achieve the desired effect.

III. Methodology

The overall architecture of the proposed DS-RDMPD is depicted in Fig. 2. The framework consists of two consecutive stages: a coarse dehazing stage based on MCESS-UNet, and a refinement stage based on RDMPD. In the first stage, MCESS-UNet enhances the visibility of hazy images through multiscale feature extraction and adaptive fusion. The output x_{pre} is concatenated with the original hazy image to form the input x_{in} for the second stage, where residuals are refined stepwise using the diffusion model.

This DS design is inspired not only by the layered degradation characteristics of remote sensing haze but also by the shared structural patterns across different image restoration tasks, such as deraining and deblurring. These tasks often exhibit compound degradations that can be decomposed into coarse structural distortions and fine residual noise. The first stage captures global degradations through hierarchical representations, while the second stage corrects fine-scale distortions with learned priors. By decoupling these two processes, the DS-RDMPD achieves strong generalization across heterogeneous restoration domains.

A. Multichannel Efficient Selective Synthesis U-Net

Considering the uneven distribution, multiscale structure, and heterogeneous texture of remote sensing haze images, as well as the large amount of computation required by traditional Transformer-based methods, we aim to use the attention mechanism to balance efficiency and effectiveness. We then designed two blocks, one for feature extraction and one for feature selection. The MC block is employed to extract

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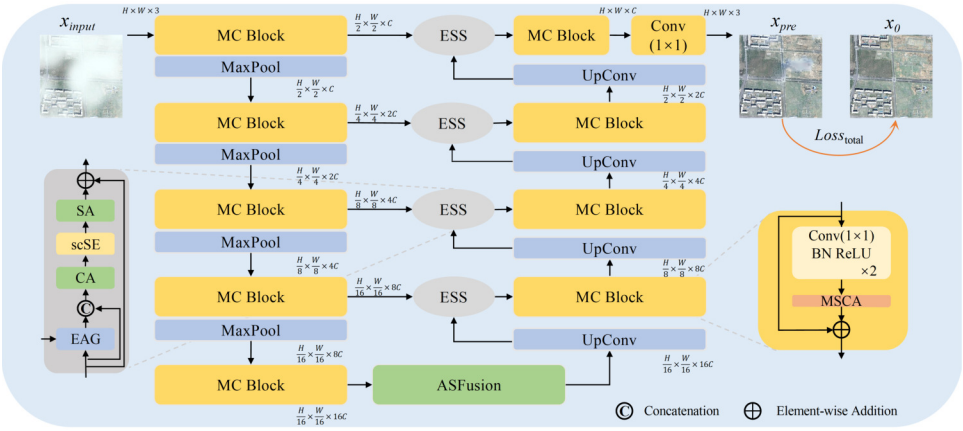


Fig. 3. Schematic of the developed MCESS-Net. We propose a multiscale feature learning network with an encoder–decoder hierarchical structure. The network starts with the input image; downsamples it four times, each time through MC block and max pooling, and the bottom layer is fused through an MC block and an ASFusion layer; and then upsamples it four times, each time through Upconv and MC block, where Upconv changes the shape and size, and then, the tensors corresponding to the upsampling and downsampling enter the ESS block and finally generate the predicted output.

features at different scales, and the ESS block subsequently applies selective weighting and fusion of these features to enhance dehazing performance.

The MCESS-UNet network is shown in Fig. 3. The encoder uses MC blocks to extract hierarchical features while reducing spatial dimensions via max pooling. In the decoder, spatial resolution is progressively reconstructed with standard convolutions. At each stage, MC blocks capture deeper features, which are fused with corresponding encoder features through ESS blocks, replacing traditional skip connections. The MC block enhances multiscale features, while the ESS block selectively preserves channel and spatial information. In Fig. 3, the lower right is the architecture diagram of the MC block, and the lower left is the architecture diagram of the ESS block. These blocks complement each other, effectively extracting multiscale features and analyzing spatial and channel information. In addition, we introduce the ASFusion module of our previous work, DeHaze-UNet [32], to further fuse features. The final output consists of the predicted image x_{pre} and an auxiliary low-resolution output x_0, both optimized using a hybrid loss function $\text{Loss}_{\text{total}}$.

1) MC Block: In traditional CNNs, the size and stride of the convolution kernel are fixed, which means that the receptive fields processed by the network at different levels are hard-coded and cannot change adaptively based on the input data. Attention mechanisms help capture these relationships, but they still have difficulty handling multilevel features. We therefore design the MC block, which combines convolution with a multiscale channel-aware (MSCA) mechanism to optimize feature selection and expressiveness.

The MC block adopts a MSCA block to improve the ability to extract features. As shown in Fig. 4, it enhances the traditional CNN architecture by combining multiscale feature maps with a channel-aware mechanism. As shown in Fig. 4, the block uses convolution kernels of different sizes (i.e., 1×1 (pixel by pixel), 3×3, and 5×5) and max pooling to capture features at different scales, integrates the outputs through 1×1 convolution, and adopts a channel-aware mechanism

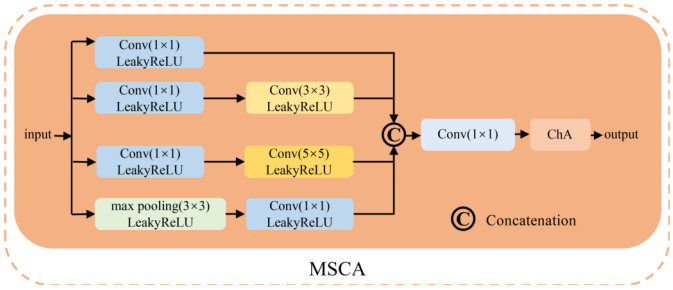


Fig. 4. Architecture of the complete MSCA. ChA is the channel-aware module.

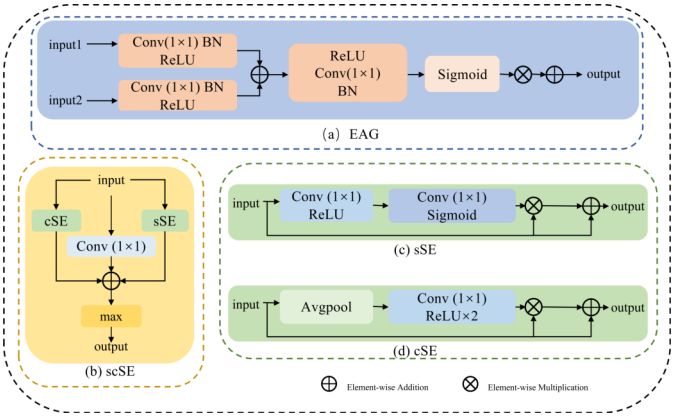


Fig. 5. (a) Internal details of the EAG structure. (b) Composition diagram of the scSE block. (c) Internal structure of sSE. (d) Internal structure of cSE.

to selectively enhance important features while suppressing irrelevant or noisy information. This enables the model to focus on the most discriminative features.

2) ESS Block: While traditional skip connections facilitate gradient flow, they are suboptimal in preserving fine-grained, multiscale features. The proposed ESS block addresses this by incorporating a series of attention modules—including an efficient attention gate (EAG), a CA mechanism, and a spatial–channel squeeze-and-excitation (scSE) block—to adaptively modulate feature maps. As shown in Fig. 5, the

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final output is further refined through spatial attention to compensate for suppressed spatial cues.

As shown in Fig. 5, ESS combines the output tensors of the downsampled MC block and the upsampled UpConv (both have corresponding shapes) into inputs 1 and 2. The two input features are first fused with global context information through an EAG and then adaptively adjusted using the CA mechanism to highlight key channel features. These features are then processed by the scSE block, which simultaneously applies channel squeeze and excitation (cSE) and spatial squeeze and excitation (sSE), where in cSE, spatial information is extracted through global average pooling, and then, CA weights are generated through a fully connected layer and a sigmoid function to recalibrate the feature map and emphasize important channels. In sSE, a 1×1 convolution is followed by a sigmoid activation function to enhance the feature map, create a spatial attention map, refine the input, and highlight important spaces. The two are added to the residual and the important max operator is used to select the eigenvalue with the largest weight. Finally, the output feature map is optimized in both channel and spatial dimensions, providing powerful representation capabilities. It is worth mentioning that we added a spatial attention at the end because the selection mechanism will suppress some ignored important features, so we emphasize that the spatial attention is combined with the residual to obtain the final output feature of ESS.

3) Loss Function: The design of the loss function is crucial for generating high-quality images. We adopt a composite loss that combines standard L1 loss, mean squared error (mse) loss, and a custom-designed L1-scale loss to optimize both pixel-level accuracy and structural fidelity. The L1-scale loss is defined as

$$\text{Loss}_{\text{L1-scale}} = \|R(y) - R(\hat{y})\|_1 \tag{1}$$

where $\hat{y}$ denotes the generated image and y is the ground truth. $R(\cdot)$ denotes a restoration to original scale operation that multiplies each pixel value by 255, mapping the output back to the original 8-bit intensity scale of remote sensing images. This design is specifically motivated by the inherent characteristics of remote sensing imagery, in which small pixel-level differences—such as weak edges or subtle haze gradients—encode crucial cues related to land cover boundaries, object contours, and surface textures. Compared to natural images, structural cues in remote sensing data often manifest as subtle intensity variations, particularly under hazy or low-contrast conditions, which are often insufficiently captured by conventional loss functions.

By amplifying the magnitude of pixel residuals, L1-scale loss increases the model's sensitivity to small but significant local variations. However, we acknowledge that oversensitivity may amplify noise or imperceptible changes. To mitigate this, we integrate L1-scale into a balanced loss function alongside L1 and mse terms. The complete formulation is

$$\text{Loss}_{\text{L1}} = \|y - \hat{y}\|_1 \tag{2}$$

$$\text{Loss}_{\text{mse}} = \|y - \hat{y}\|_2^2 \tag{3}$$

$$\text{Loss}_{\text{total}} = \text{Loss}_{\text{L1}} + \text{Loss}_{\text{mse}} + \lambda \cdot \text{Loss}_{\text{L1-scale}} \tag{4}$$

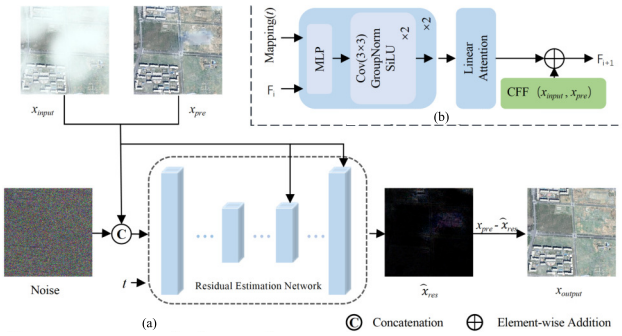


Fig. 6. Proposed RDMPD with perceptual decoding (a) represents the sampling process. By iteratively embedding the two conditions (x_{input} and x_{pre}) along with the noise in the mapping of time t, the residual is gradually estimated and denoised to obtain the predicted residual result, x_{pre} minus $\hat{x}_{\text{res}}$ to get the output. (b) Perceptual decoding part, where the features of x_{input} and x_{pre} are decoupled by the CFF block.

where λ is empirically set to 2 to ensure a proper balance across loss terms. This hybrid loss maintains overall reconstruction fidelity while better preserving fine haze-affected structures common in remote sensing scenarios.

B. Residual Diffusion Model With Perceptual Decoding

RDDM [14] is a new double diffusion process. To further improve the accuracy of model generation, we design a CFF module to decouple conditions, which we call perceptual decoding, aiming to further utilize conditional reference textures to guide the estimation residual and denoising process. The proposed method is shown in Fig. 6. During training, x_{in} is the image that x_{pre} and hazy depend on, where hazy is an additional condition to prevent the model from being too dependent on the results of the first stage and enhance the robustness of the model. Next, we describe the forward process, sampling process, and perceptual decoding process.

1) Forward Process: The forward diffusion process progressively transforms a clean image into pure noise by iteratively adding the structural residual and stochastic noise. We define the residual term x_{res}, which represents the difference between the clear image and the preprocessed image: $x_{\text{res}} = x_{\text{pre}} - x_0$, where x_0 is the clear image and x_{pre} is the preprocessed image generated by the pretrained MCESS-UNet. The residual x_{res} provides directional guidance in the forward process, encoding the structural differences between the clear image and the preprocessed one. In our forward process, each time step adds both residuals and noise

$$x_t = x_{t-1} + \alpha_t x_{\text{res}} + \beta_t \epsilon_{t-1}. \tag{5}$$

By recursively expanding (5), we obtain

$$x_t = x_0 + \sum_{i=1}^{t} \alpha_i x_{\text{res}} + \sqrt{\sum_{i=1}^{t} \beta_i^2} \cdot \epsilon \tag{6}$$

where x_t is the image at time step t and α_t controls the contribution of the residual x_{res} at each time step. β_t controls the contribution of noise ϵ, which is sampled from a standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Since ϵ_t is sampled from a Gaussian distribution, x_t follows the Gaussian distribution given

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above. The conditional distribution of x_t given x_0 and x_{res} is defined as

$$q(x_t|x_0, x_{\text{res}}) = \mathcal{N}\left(x_t; x_0 + \sum_{i=1}^{t} \alpha_t x_{\text{res}}, \sum_{i=1}^{t} \beta_t^2 \mathbf{I}\right). \quad (7)$$

This recursive formula enables image generation at any given time step t. The model is optimized by minimizing the absolute difference between the predicted residuals and the true residuals so that it can accurately predict the residuals. The simplified loss function is shown as follows:

$$L_{\text{res}} = \mathbb{E}\left[\lambda_{\text{res}} \|x_{\text{res}} - \hat{x}_{\text{res}}(x_t, t, x_{\text{in}})\|_1\right] \quad (8)$$

where x_{res} represents the true residual and $\hat{x}_{\text{res}}$ is the residual predicted by the model. The parameter λ_{res} in our strategy is 1. It is worth mentioning that to enhance robustness and better preserve the structural details of the image, we use the L1 loss, which can suppress the influence of outliers to some extent. We replace the squared L2 norm in the original loss function in RDDM [14] with the L1 norm. To obtain a more sparse prediction residual with sharper edges

$$\epsilon = \frac{x_t - x_0 - \sum_{i=1}^{t} \alpha_i x_{\text{res}}}{\sqrt{\sum_{i=1}^{t} \beta_i^2}}. \quad (9)$$

In addition, the noise can be derived from the predicted residuals by the following formula, which is derived from (6). This formulation preserves the independence and Gaussianity of the noise, consistent with the principles of diffusion-based modeling.

2) Sampling Process: As shown in Fig. 2, the reverse diffusion process progressively removes noise from the x_T state to restore a clear image. Since the strategy we chose is to estimate only the residual, our reverse process is to use the trained network model to estimate the residual. The inverse process is derived from (5) as follows:

$$x_{t-1} = x_t - \alpha_t \hat{x}_{\text{res}} - \beta_t \hat{\epsilon}_{t-1} \quad (10)$$

where $\hat{x}_{\text{res}}$ is the residual predicted by the U-Net in the diffusion model and $\hat{\epsilon}_{t-1}$ is derived from (9). When sampling backward, the residual is predicted first, and then, the noise is gradually removed by adjusting the step size

$$x_0 = x_1 - \alpha_1 \hat{x}_{\text{res}}. \quad (11)$$

Equation (11) is derived from (10). When t equals 0, the final denoised image is generated. For a more detailed derivation process, please refer to RDDM [14].

3) Perceptual Decoding: Traditional diffusion models typically use basic upsampling operations for decoding, leading to a heavy reliance on encoder output features and causing the loss of important information. This approach has limitations in restoring image details. To address these issues and enhance image restoration quality, we introduced the CFF module within the encoder. The CFF module decouples the conditional information in the image, offering more structured guidance to the decoder through interpretable dynamic conditional perception.

As shown in Fig. 6(b), we integrate the features from the x_{input} and x_{pre} images into the perceptual encoder after the CFF operation. Specifically, we process the features from both inputs

$$\begin{aligned} F_p &= \text{Conv}_{3\times3}\left(\text{LeakyRelu}\left(\text{Conv}_{3\times3}\left(x_{\text{pre}}\right)\right)\right) \\ F_{\text{in}} &= \text{Conv}_{3\times3}(\text{LeakyRelu}\left(\text{Conv}_{3\times3}\left(x_{\text{input}}\right)\right)) \end{aligned} \quad (12)$$

where $\text{Conv}_{3\times3}$ represents the convolution kernel 3×3 convolution and LeakRelu is the activation function. Through such operations, we get our conditions F_p and F_{in}. We concatenate these two features. Then, we concatenate the two features together

$$F_c = \text{Concatenation}\left(F_p, F_{\text{in}}\right). \quad (13)$$

The concatenated features are represented as F_c. Then, we further learn the spatial adaptive weights and biases as follows:

$$F_i = \omega_i(F_c) \cdot F_p + \gamma_i(F_c) \quad (14)$$

where $\omega_i(\cdot)$ and $\gamma_i(\cdot)$ are the spatially adaptive weights and biases learned from the combination of the previously obtained guiding features F_c and F_p, respectively, and i indicates that this is the feature information of the ith layer of the decoder. This improvement not only enhances the perception ability of the encoder but also significantly improves the ability of the decoder to restore details.

IV. Experiments

A. Experimental Settings

1) Databases: We evaluated our DS-RDMPD on eight public datasets, each with different characteristics and uses, including six remote sensing haze datasets, one raindrop dataset, and one blur dataset. The StateHaze1k [33] dataset contains three subsets: thin haze, moderate haze, and thick haze. We merge these three subsets together, with a total of 900 pairs of images for training, and 45 pairs of images in each subset for testing. RSID [7] is a dataset for military sites containing 1000 pairs of images, each with a corresponding version. We use a stratified random method to select 100 pairs of images for testing. Specifically, we select one pair of images from each group of ten images, for a total of 100 groups, and the remaining 900 pairs of images are used for training. The DHID [36] dataset is designed for the dense haze dehazing task, contains 14 490 pairs of images for training, and has a separate 500 pairs of test sets. The LHID [36] dataset is built for synthetic optical dehazing and contains a training set and two independent test sets: LHID-A [36] and LHID-B [36]. This study uses LHID-A [36] for testing. RainDrop [37] is shot through two identical pieces of glass: one is sprayed with water and the other remains clean. There are 861 training images and 58 test images for image raindrop removal. Realblur [38] is a real-world blur dataset for learning and evaluating deblurring algorithms, with 3843 pairs of training images and 1019 test images. The above image sizes are all 512×512 (except RSID [7] and RainDrop [37], where the RSID image size is 256×256 and RainDrop is 720×480).

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TABLE I

STATEHAZE1K [33] AND RSID [7] DATASETS REMOTE SENSING IMAGE DEHAZING EFFECTS. BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST QUANTITATIVE PERFORMANCE, RESPECTIVELY

Methods	Venue	Thin		Moderate		Thick		Average		RSID	
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DCP [2]	CVPR'09	20.47	0.8705	20.83	0.8971	16.71	0.7435	19.34	0.8370	17.59	0.8488
AOD-Net [4]	ICCV'17	16.36	0.8168	17.88	0.8666	15.87	0.7402	16.70	0.8079	18.78	0.8579
FFA-Net [34]	AAAI'20	25.16	0.9168	24.79	0.9296	22.36	0.8453	24.10	0.8972	26.04	0.9519
Trinity-Net [7]	TGRS'23	23.46	0.8998	25.98	0.9372	20.47	0.8148	23.30	0.8839	25.34	0.9455
Dehazeformer-b [6]	TIP'23	25.68	**0.9193**	27.15	0.9408	22.74	**0.8494**	25.19	0.9032	25.79	0.9548
OK-Net [35]	AAAI'24	22.16	0.8974	26.00	0.9437	20.51	0.8257	22.89	0.8889	25.02	0.9489
AU-Net [29]	TGRS'24	22.71	0.8992	21.04	0.9038	21.81	0.8458	21.85	0.8829	17.01	0.7671
DS-RDMPD	-	**26.03**	0.9191	**28.17**	**0.9444**	**22.97**	0.8485	**25.72**	**0.9040**	**26.32**	**0.9548**

2) Experimental Setup: Our model consists of two stages. In the first stage, MCESS-UNet is trained on two NVIDIA RTX A5000 GPUs. The model is trained for 300 epochs on StateHaze1K [33], RSID [7], RainDrop [37], and Realblur [38] datasets, and 100 epochs on DHID [36] and LHID [36] datasets, with a batch size of 6. We use the Adam optimizer with $\beta 1 = 0.9$, $\beta 2 = 0.999$, and an initial learning rate of 0.0002, which is decayed to 0.000001 by a cosine annealing scheduler. In the second stage, we follow the experimental settings in the original RDDM [14] paper and train RDMPD for 300 000 iterations (full training takes 2894.5 min for 512 $\times$ 512 size and only 500 min for 256 $\times$ 256 size) using NVIDIA RTX A5000 GPUs with a batch size of 1 and a fixed learning rate of 0.00008. We adopt the step-by-step skipping strategy of DDIM [9] for sampling, and the final result is obtained after five steps of sampling. The DS-RDMPD inference time is 0.4619 s for 256 $\times$ 256 images and 1.5752 s for 512 $\times$ 512 images.

B. Comparison With State-of-the-Art Methods

To ensure fairness, the comparative experiments in this article are reproduced on our devices using the weights or parameter settings in the original paper, and the superiority of our model is verified from a quantitative and qualitative perspective. All tests are performed on a single NVIDIA RTX A5000 GPU. It is worth mentioning that for RSHazeDiff [10], our device cannot reproduce its model well, so we directly used the data in the original paper. Their LHID is different from our test set, so it is not shown.

1) Remote Sensing Dehazing Small Dataset: To test the performance of our proposed DS-RDMPD method in dehazing small synthetic remote sensing images, we conduct comparative experiments with several recognized state-of-the-art methods: DCP [2], AOD [4], FFA-Net [34], Trinity-Net [7], Dehazeformer-b [6], OK-Net [35], and AU-Net [29], and present the quantitative results on StateHaze1K [33] and RSID [7] datasets. The bold numbers in Table I are the best data, and the underlined numbers are the suboptimal data. As shown in Table I, our DS-RDMPD achieves the highest PSNR scores across all subsets and yields the best SSIM scores in most cases. Notably, Dehazeformer-b [6] achieves slightly higher SSIM scores than ours on the StateHaze1K-thin and StateHaze1K-thick datasets by only 0.0002 and 0.0009, respectively. It is worth emphasizing that our model outperforms all others on the StateHaze1K-moderate subset, achieving a PSNR that is 1.02 dB higher than the second-best model, Dehazeformer-b.

We also conduct qualitative evaluations on these two datasets. Specifically, the three rows of Fig. 7 show the effects of three different levels of haze in StateHaze1K [33]: thin, moderate, and thick. For the thin images in the first row, DCP, AOD-Net, and AU-Net all have color distortion. FFA-Net, Trinity-Net, and OK-Net are all affected by haze to varying degrees. The enlarged blue block is a bit white, and the upper left land of Dehazeformer-b is a bit whitish. Our model obtained pleasant results. For the second row of modified images, the enlarged image of Trinity-Net obviously has no other color shift compared to other models, but it is not restored to the original color, and the natural restoration of the land is very unnatural. The green grass is dark in color. Our results not only closely resemble the natural appearance of the ground truth but also achieve a PSNR score that is 1.97 dB higher than that of the second-best method. For the thick images in the third row, our images restore the ground truth well compared to other models. Although the color may be a bit dark, other models are affected by severe haze and either have a serious color shift or fail to restore details.

Similarly, Fig. 8 describes the visual effect comparison of RSID [7] using different dehazing methods. It can be observed that the images restored by DCP and AOD retain residual haze. The color distortion of the image restored by AU-Net is too serious. In the enlarged image, the yellow object in the image restored by FFA-Net is obviously too bright, the yellow object in the image restored by Trinity-Net is dark, the yellow block in the image restored by OK-Net has obvious color distortion, and the sea surface is too dark. In the enlarged image, the color of the deck of Dehazefoemer-b is too dark, and the color restoration of the clear image is not good. Our DS-RDMPD removes haze effectively, accurately restores the colors of the yellow object and the deck in the enlarged region, and surpasses the second-best method by 2.06 dB in PSNR and 0.0049 in SSIM.

2) Remote Sensing Dehazing Large Dataset: On large-scale datasets such as DHID [36] and LHID [36], we compare with DCP [2], AOD [4], FCTF-Net [39], GridDehaze-Net [40], FFA-Net [34], MixDehaze-Net [41], AU-Net [29], and RSHazeDiff [10], and the quantitative results are summarized in Table II. As shown in Table II, our DS-RDMPD achieves the

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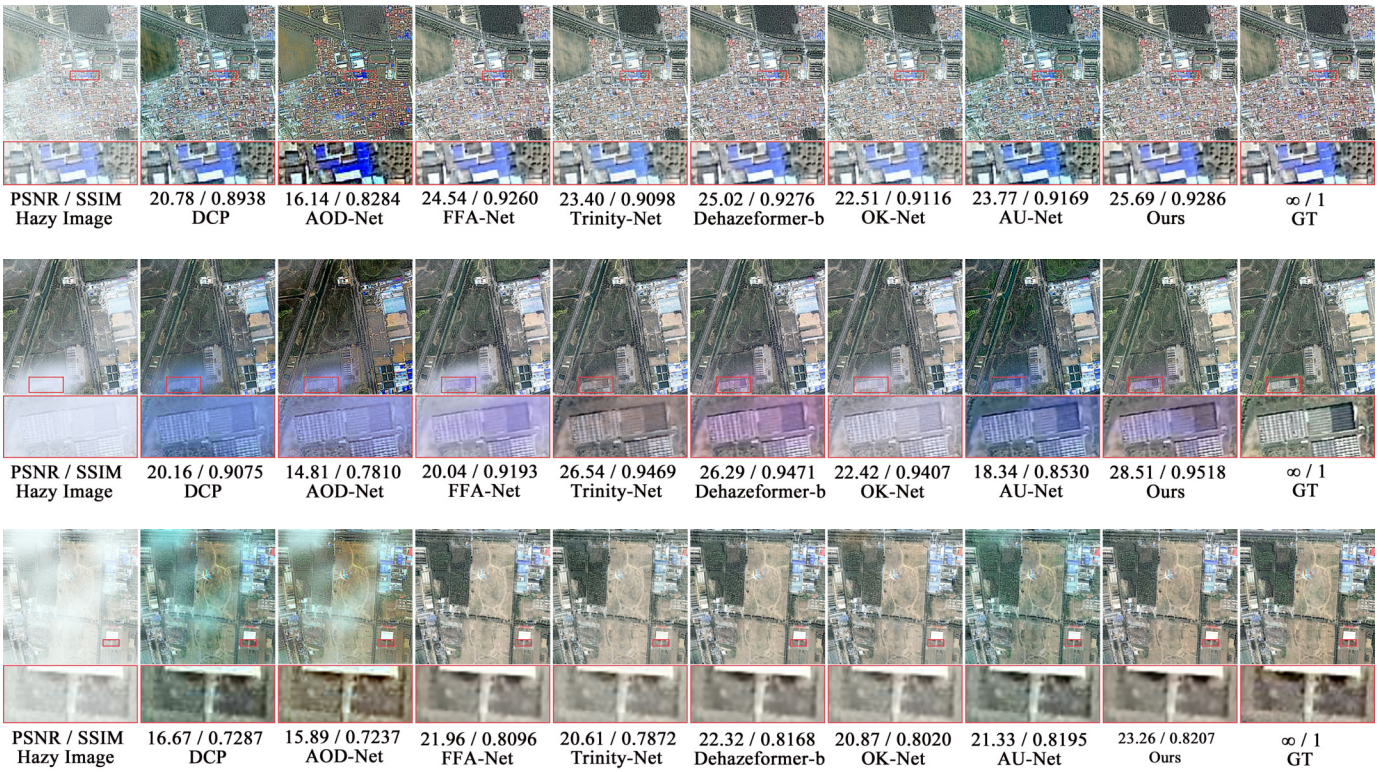


Fig. 7. Visual samples generated by the evaluation method on the StateHaze1K [33] dataset. From the first row to the third row, they are the restoration of haze images of three subsets: thin, moderate, and thick.

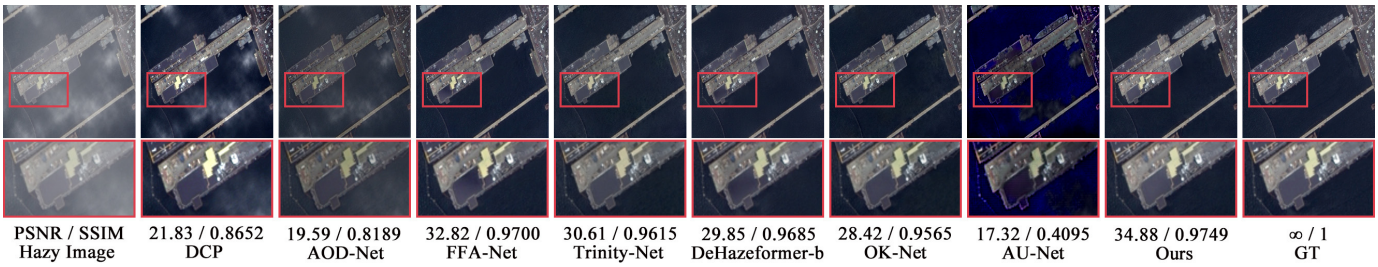


Fig. 8. Visual samples generated by the evaluated methods on the RSID [7] dataset.

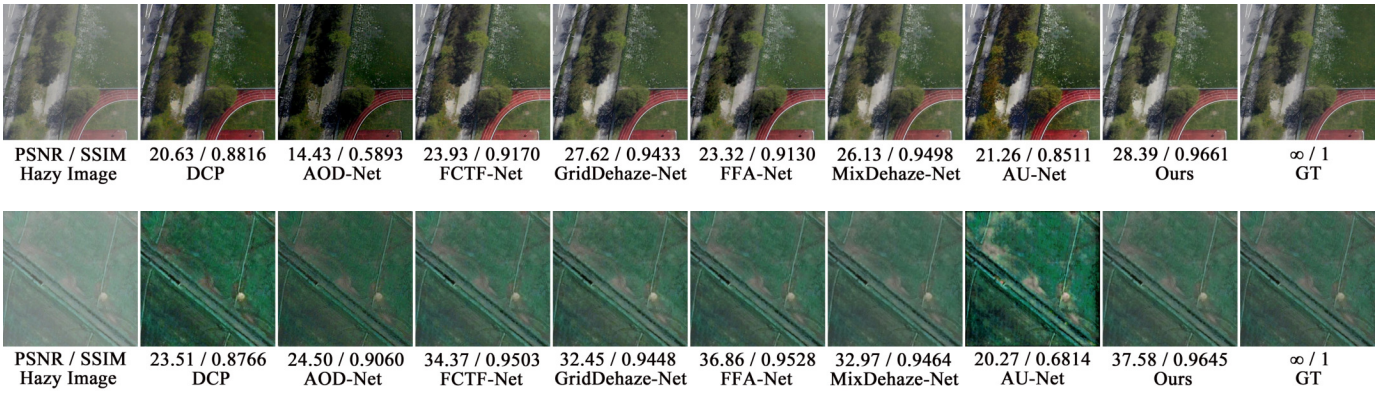


Fig. 9. Visual samples generated by the evaluation method on the DHID [36] and LHID [36] datasets, where the first row is DHID and the second row is LHID.

highest PSNR and SSIM scores on the LHID dataset, with the SSIM score exceeding that of the second-best method, FFA-Net, by 0.0099.

Fig. 9 illustrates a visual comparison of different methods on the LHID and DHID datasets. The first row presents the results of various models on the DHID dataset. It can be observed that

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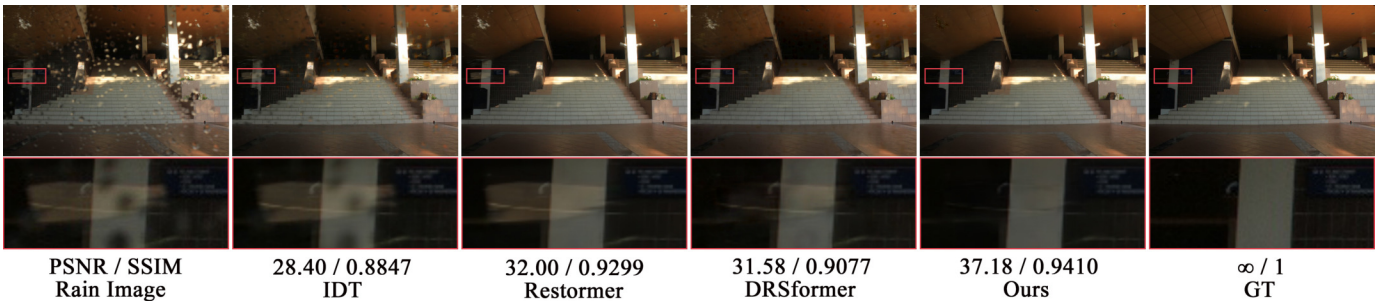


Fig. 10. Visual samples generated by the evaluated methods on the RainDrop [37] dataset.

TABLE II

LHID [36] AND DHID [36] DATASETS REMOTE SENSING IMAGE DEHAZING EFFECTS. BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST QUANTITATIVE PERFORMANCE, RESPECTIVELY

Methods	Venue	LHID		DHID	
		PSNR	SSIM	PSNR	SSIM
DCP [2]	CVPR'09	21.34	0.7975	19.15	0.8194
AOD-Net [4]	ICCV'17	21.91	0.8144	16.03	0.7291
FCTF-Net [39]	GRSL'21	28.55	0.8727	22.43	0.8482
GridDehaze-Net [40]	ICCV'19	25.80	0.8584	26.77	0.8851
FFA-Net [34]	AAAI'20	29.33	0.8755	24.62	0.8657
MixDehaze-Net [41]	IGCNN'24	29.37	0.8624	27.25	0.8844
AU-Net [29]	TGRS'24	16.82	0.5993	23.73	0.8573
RSHazeDiff [10]	TITS'24	-	-	27.91	0.9000
DS-RDMPD	-	**29.39**	**0.8854**	**27.91**	**0.9054**

TABLE III

REAL-WORLD RAINDROP REMOVAL EFFECT ON THE RAINDROP [37] DATASET. BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST QUANTITATIVE PERFORMANCE, RESPECTIVELY

Methods	Venue	RainDrop	
		PSNR	SSIM
IDT [42]	TPAMI'22	28.33	0.8993
Restormer [43]	CVPR'22	**31.06**	**0.9242**
DRSformer [44]	CVPR'23	30.11	0.9139
DS-RDMPD	-	30.22	0.9154

the restored images of the DCP, AOD-Net, and AU-Net models are too dark in color, while the restored red playground runway in the GridDehaze-Net and FFA-Net results are too bright. In addition, the haze in the upper right corner of the FCTF-Net and MixDehaze-Net results is not completely removed. Our DS-RDMPD removes the haze very cleanly overall, restores the color of the playground runway very well, and achieves the highest scores in both PSNR and SSIM indicators. The second row is a visualization comparison of the models on the LHID dataset, where the results of DCP, AOD-Net, and AU-Net have obvious color deviations, and the ground truth is not fully reflected. Although there is no significant visual difference compared to other methods, our model achieves the highest PSNR and SSIM scores, surpassing the second-best method, FFA-Net, by 0.72 dB in PSNR and 0.0117 in SSIM.

3) Raindrop Image Removal Dataset: In the raindrop removal task, we selected the RainDrop [37] dataset to verify the raindrop removal ability of our model. We selected several state-of-the-art models for comparison. Specifically, we compared with IDT [42], Restormer [43], and DRSformer [44]. The results are summarized in Table III. As can be seen from Table III, on the raindrop removal dataset, Restormer achieved the best score, while our DS-RDMPD achieved the second-best PSNR and SSIM scores. Compared with these models dedicated to rain removal tasks, our model also showed satisfactory results, which is enough to prove the potential and excellent generalization ability of DS-RDMPD in image restoration tasks.

Fig. 10 shows the visual comparison of different raindrop removal methods on the RainDrop dataset. It can be observed that the results of the IDT model clearly have raindrop residues. The zoomed-in results of Restormer and DRSformer show that the light pollution reflected by raindrops cannot be removed. Our method effectively removes light pollution, delivers satisfactory visual results, and achieves the highest PSNR and SSIM scores, with a 5.18-dB improvement in PSNR and a 0.0111 improvement in SSIM over the second-best method, Restormer.

TABLE IV

REAL-WORLD DEBLURRING EFFECT ON THE REALBLUR [38] DATASET. BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST QUANTITATIVE PERFORMANCE, RESPECTIVELY

Methods	Venue	Realblur	
		PSNR	SSIM
Uformer [45]	CVPR'22	30.11	0.9057
NAFNet [46]	ECCV'22	30.09	0.8939
MLWNet [47]	CVPR'24	30.27	**0.9082**
DS-RDMPD	-	**30.50**	0.8981

4) Deblurred Image Dataset: In the deblurring task, we selected the Realblur [38] dataset to verify the deblurring effect of our model. We selected several state-of-the-art models for comparison. Specifically, we compared with Uformer [45], NAFNet [46], and MLWNet [47]. The results are summarized in Table IV. As shown in Table IV, on the deblurring dataset, our DS-RDMPD achieved the highest PSNR, outperforming the second-best model, MLWNet, by 0.23 dB. Although MLWNet obtained the highest SSIM, our model's SSIM was only 0.0101 lower. Despite not being optimal in SSIM, these results demonstrate the strong potential and excellent generalization ability of DS-RDMPD in image restoration tasks.

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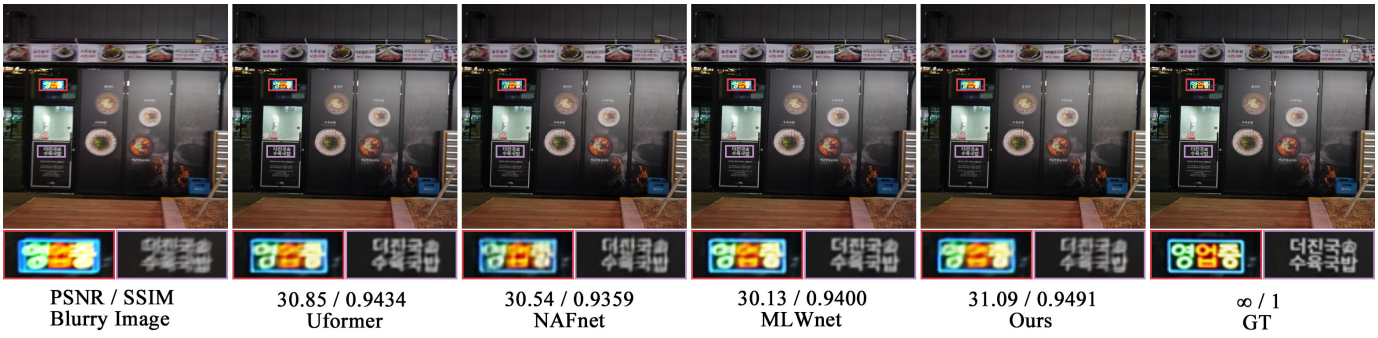


Fig. 11. Visual samples generated by the evaluated methods on the Realblur [38] dataset.

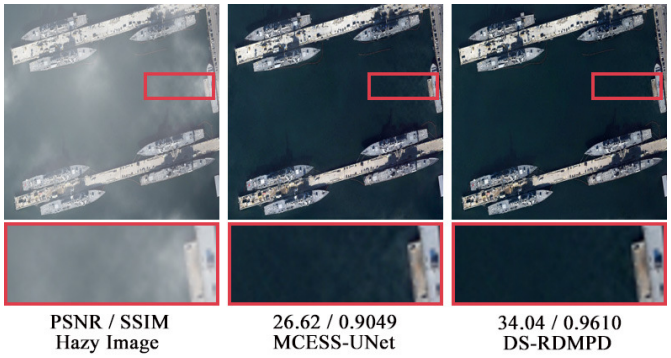


Fig. 12. (Left to right) Results of the MCESS-UNet model in stage 1, the results of the DS-RDMPD in stage 2, and the original clear image (image in the RSID [7] dataset).

Fig. 11 shows the visual comparison of different deblurring methods on the Realblur [38] dataset. Other models, including ours, also have obvious deblurring effects. Although the specific gap is not obvious, our model achieved the highest scores in both PSNR and SSIM, surpassing the second-best model, Uformer, by 0.24 dB in PSNR and 0.0057 in SSIM.

C. Ablation Study

We first show the output images of the two stages for comparison, as shown in Fig. 12. The results of the first-stage MCESS-UNet obviously fail to remove the haze cleanly, while the second-stage DS-RDMPD achieves satisfactory results. However, it is precisely because the haze in the first stage has less impact that our diffusion model is able to quickly remove the remaining haze, which greatly reduces our training resources and only requires about five sampling steps to obtain satisfactory results. The following are the ablation experiments in the DS experiment.

1) Ablation Study of MCESS-UNet: Table V shows the results of 200 rounds of training on the Haze1K [33] dataset and subsequent testing on thin images. All training parameters such as learning rate, batch size, loss function, and devices, are kept constant for comparison purposes. The table highlights the main contributions of the two blocks introduced in this article: the MSCA block and the ESS block. Specifically, the MSCA block and the ESS block can significantly improve PSNR and SSIM when used alone. Compared to the baseline, the ESS block improves PSNR by 0.86 dB and SSIM by 0.0074, while the MSCA block increases PSNR by 1.19 dB and SSIM by 0.0091. When the two blocks are used together, their combined effect can significantly improve the performance, achieving a total gain of 2.02 dB in PSNR and 0.0173 in SSIM over the baseline.

TABLE V

RESULTS OF THE ABLATION STUDY FOR THE MSCA BLOCK AND ESS BLOCK. BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST QUANTITATIVE PERFORMANCE, RESPECTIVELY

Model	PSNR	SSIM
Baseline Model	23.05	0.8999
Model with ESS	23.91	0.9073
Model with MSCA	<u>24.24</u>	<u>0.9090</u>
Model with MACA+ESS	**25.07**	**0.9172**

TABLE VI

RESULTS OF A COMPARATIVE EXPERIMENT ON DIFFERENT LOSS FUNCTIONS. BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST QUANTITATIVE PERFORMANCE, RESPECTIVELY

Loss-type	PSNR	SSIM
L1	24.56	0.9122
MSE	24.58	0.9105
L1-scale	24.14	0.9114
L1+MSE	24.93	<u>0.9153</u>
MSE+L1-scale	24.77	0.9121
L1+L1-scale	<u>24.97</u>	0.9140
L1+MES+L1-scale	**25.07**	**0.9172**

Table VI presents the results of different loss function configurations. Among the single loss settings, L1 achieves relatively good performance (24.56 PSNR/0.9122 SSIM), while L1-scale and mse perform worse when used alone. In particular, L1-scale loss alone results in the lowest PSNR (24.14), suggesting that although it enhances sensitivity to pixel-level differences, it may amplify noise or unstable gradients when not combined with other terms. This is consistent with its design motivation—to emphasize subtle haze variations and fine textures common in remote sensing images.

When the L1-scale is combined with L1 or mse, both PSNR and SSIM show clear improvements. For instance, L1 + L1-scale reaches 24.97 PSNR, slightly higher than L1 + mse (24.93). The best results are achieved when all three losses are used together, yielding 25.07 PSNR and 0.9172 SSIM. This

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TABLE VII

EFFECTS OF DIFFERENT COMBINATIONS OF CONDITIONS AND PERCEPTUAL DECODERS ON MODEL PERFORMANCE. BOLD AND UNDERLINE REPRESENT THE BEST AND SECOND-BEST QUANTITATIVE PERFORMANCE, RESPECTIVELY

Model	Condition	Input Condition	PSNR	SSIM
RDDM	x_{hazy}	-	24.53	0.9059
	x_{pre}	-	25.56	0.9179
	x_{hazy}	x_{pre}	25.37	0.9086
	x_{pre}	x_{hazy}	<u>25.79</u>	<u>0.9188</u>
DS-RDMPD	x_{pre}	x_{hazy}	**26.03**	**0.9191**

confirms that the three losses are complementary; together, they form a balanced and effective objective for remote sensing image dehazing.

2) Ablation Study of RDM: This section introduces the modifications to the input conditions of RDDM [14] and the effectiveness of our perceptual decoder. First, we adjust the condition and input condition settings while keeping all other parameters unchanged to verify the superiority of our design. The residual in RDDM is calculated as $x_{\text{condition}} - x_0$, where the "condition" refers to the image used to guide the residual prediction and the "input condition" refers to the image used as an auxiliary input for the denoising network. The input condition does not participate in the residual calculation.

Table VII summarizes the impact of different combinations of condition and input condition on model performance. The first four rows correspond to the RDDM baseline, where no perceptual decoder is involved. When using only x_{hazy} as the condition, the model achieves relatively low performance (24.53 PSNR/0.9059 SSIM). Replacing the condition with the intermediate result x_{pre} increases the PSNR to 25.56 and SSIM to 0.9179, indicating that cleaner guidance helps produce better residual estimation.

Furthermore, when x_{pre} is used as the condition and x_{hazy} is used as the input condition, the model achieves even better results: 25.79 in PSNR and 0.9188 in SSIM, outperforming the pure x_{pre} input by 0.23 dB and 0.0009, respectively. This demonstrates that although x_{hazy} is a degraded observation, it still provides valuable structural information to enhance robustness and performance.

The last row corresponds to DS-RDMPD, which incorporates a perceptual decoder. With the same conditioning strategy as row 4, the perceptual decoder further boosts the performance to 26.03 in PSNR and 0.9191 in SSIM. Compared to the best RDDM variant, this reflects an additional improvement of 0.24 dB in PSNR and 0.0003 in SSIM, demonstrating the effectiveness of our perceptual decoding module.

These results confirm that our condition design, together with the perceptual decoder, contributes significantly to the performance improvements in remote sensing image dehazing.

V. CONCLUSION

This article introduces DS-RDMPD. First, a pretrained MCESS-UNet is used for preprocessing, and the resulting image is then used to guide RDMPD in refinement. By integrating MCESS-UNet with RDMPD, this DS framework improves image quality while reducing computational load, striking a balance between expensive training costs and actual recovery effectiveness. In addition, DS-RDMPD not only shows great potential in the field of remote sensing dehazing but is also effective in tasks such as deraining and deblurring, highlighting its strong generalization across various image restoration problems. Future work aims to optimize the model to make it more suitable for practical applications and enhance its robustness and scalability to complete more tasks, such as dehazing in water and desnowing in real scenes. However, the current model still faces limitations in handling high-resolution images, where restoring fine details requires more advanced architectures and efficient memory management.

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